Chromosomes, Genes & Proteins

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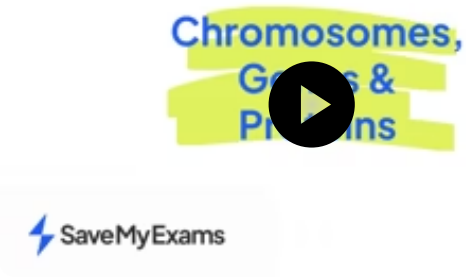

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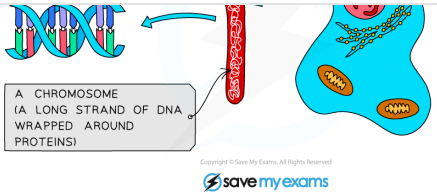
The Structure of Genes

Inheritance of Chromosomes, Genes and DNA

- **Inheritance** is the **transmission of genetic information from generation to generation**
- **Chromosomes** are located in the **nucleus** of cells
- They are **thread-like structures of DNA, carrying genetic information in the form of genes**
- A **gene** is a short length of **DNA** found on a chromosome that **codes for a specific protein**
- This could be a structural protein such as collagen found in skin cells, an enzyme or a hormone
- Genes control our characteristics as they code for proteins that play important roles in what our cells do
- Think about zooming into the nucleus of a cell, as shown in the diagram below
  - On the right is the zoomed-out view
  - Which zooms in as we read across the diagram right-to-left

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



Genes are short lengths of DNA that code for a protein. They are found on chromosomes

- Alleles are **variations** of the same gene
  - As we have two copies of each chromosome (one from each parent), we have **two copies of each gene** and therefore **two alleles for each gene**
  - One of the alleles is inherited from the mother and the other from the father
  - This means that the alleles do not code for the same thing
  - For example, an individual has two copies of the gene for eye colour but **one allele could code for brown eyes and one allele could code for blue eyes**

Examiner Tips and Tricks

The Zoom model is always useful when trying to visualise where you are in a cell, and what molecules are doing what. Imagine a zoom lens that has the power to zoom right out to look at a whole organism, but also to zoom right into individual molecules. This helps in recognising the structures in cell nuclei visible as chromosomes, then zooming in to picture the individual genes and then, the bases that make up the DNA

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